Mitchell's Completeness Theorem: no exact polynomial invariant selector in the flux landscape

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Abstract

We prove that the D3 tadpole $N_{\text{flux}} = F^\top JH$ is the *unique* duality-invariant bilinear form of the type IIB flux under the full duality group $\mathrm{Sp}(b_3) \times \mathrm{SL}(2,\mathbb{Z})$: the ring of duality-invariant polynomials in the flux pair is generated by the tadpole alone (the First Fundamental Theorem of invariant theory, Weyl 1939, at $k = 2$) — there is no invariant of any degree other than N_{flux} and its powers. Over the reals the duality group acts transitively on each non-zero tadpole level set; over the integer group $\mathrm{Sp}(b_3,\mathbb{Z})$ the orbits are instead classified by the content $d = \gcd(F, H)$ ($d^2 \mid N$), giving *finitely many* orbits per tadpole. Consequence — a *no-fingerprint no-go* on exact selectors: no exact, polynomial, duality-invariant function of the flux can single out a vacuum. This rules out *exact polynomial invariant* selectors; it does not exclude statistical, moduli-dependent, or non-polynomial selection. The exact content of a theory of everything is constrained to two classes — consistency constraints and topological labels — and nothing else. The statement is falsifiable: a single moduli-independent, duality-invariant flux quantity that is not a function of $F^\top JH$ breaks it. All claims are verified symbolically (sympy, zero tolerance) and reproduced by `scripts/validate_all_theorems.py`.

1 Introduction

The constraint-first program measured, on 174 supersymmetric IIB flux vacua, that the low-energy observables of a flux vacuum form a mostly-unstructured cloud: the string coupling, the gauge couplings, the mass-spectrum shape, the axions, the dilaton and the cosmological constant are statistically uniform in flux, while exactly one quantity — the D3 tadpole — is exactly structured [1]. That was a measurement. This paper upgrades its central claim — *"the landscape's only exact structure is what it forbids"* — from an empirical pattern to a theorem, by identifying the flux pair as the $k = 2$ case of a classical result in invariant theory.

2 Setup

Let X be a Calabi–Yau threefold with middle cohomology $H^3(X,\mathbb{Z}) \simeq \mathbb{Z}^{b_3}$, and let the two NS–NS/RR three-form fluxes be $F, H \in H^3(X,\mathbb{Z})$. The intersection form is the antisymmetric $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ in a symplectic basis, and the D3 tadpole is the symplectic pairing

$$N_{\text{flux}} \;=\; F^\top JH \;=\; \sum_{i=0}^{b_3/2-1} \left(F_i H_{i+b_3/2} - F_{i+b_3/2} H_i\right). \tag{1}$$

The duality group is $\mathrm{Sp}(b_3,\mathbb{Z})$ (the monodromy group acting on H^3, hence on F and H separately) together with S-duality $\mathrm{SL}(2,\mathbb{Z})$ acting as $(F, H) \mapsto (aF + bH, cF + dH)$, $ad - bc = 1$. The tadpole is invariant under both: under $\mathrm{Sp}(b_3)$ it is the symplectic pairing, and under $\mathrm{SL}(2,\mathbb{Z})$ one computes $N_{\text{flux}} \mapsto (ad - bc) N_{\text{flux}}$ because $F^\top JF = H^\top JH = 0$ and $H^\top JF = -F^\top JH$.

3 Mitchell's Uniqueness Theorem

Theorem 1 (Mitchell)**.** *The D3 tadpole $F^\top JH$ is the unique non-trivial bilinear form in the flux (F, H) that is invariant under $\mathrm{Sp}(b_3, \mathbb{Z})$, for any b_3.*

Remark 1 (Scope: one symplectic pair)**.** *The statement is about* one *flux pair. With k independent pairs the invariant bilinear forms are the k^2 pairings $F_i^\top JH_j$: a direct computation over the full symplectic Lie algebra gives exactly one invariant for $k = 1$ (at $b_3 = 2, 4, 6$) and exactly four for $k = 2$. A compactification with two independent flux sectors therefore has four exact observables rather than one, and any claim of uniqueness must name the sector structure it assumes.*

The candidates $F^\top F$, $H^\top H$, $F^\top H$ are gauge-dependent (they change under a symplectic transformation), while $F^\top JF = H^\top JH = 0$ identically by antisymmetry. The remaining pairing is $F^\top JH$. This is verified symbolically (6/6 checks, `scripts/validate_mitchell_uniqueness.py`).

4 The Constraint-First Theorem: the full invariant ring

Mitchell's theorem is the degree-2 slice. The whole curve is classical.

Theorem 2 (Constraint-First, = First Fundamental Theorem, Weyl 1939)**.** *Every $\mathrm{Sp}(b_3)$-invariant polynomial in the flux pair (F, H) is a polynomial in the tadpole:*

$$\mathbb{C}\big[(\mathbb{C}^{b_3})^2\big]^{\mathrm{Sp}(b_3)} \;=\; \mathbb{C}\big[F^\top JH\big]. \tag{2}$$

This is Weyl's First Fundamental Theorem of invariant theory for the symplectic group specialized to $k = 2$ copies of the defining representation; the flux pair $(F, H) \in \mathbb{C}^{b_3} \oplus \mathbb{C}^{b_3}$ is precisely that $k = 2$ case. It says the tadpole *generates* the invariant ring: the only invariants are $1,\ F^\top JH,\ (F^\top JH)^2,\ \ldots$ We verify the completeness direction computationally at every degree $d = 1, \ldots, 5$ by annihilating a torus-reduced ansatz with the full $\mathfrak{sp}(b_3)$ algebra and computing the nullity exactly; the invariant-space dimensions are

$$\dim = 0,\ 1,\ 0,\ 1,\ 0 \qquad (d = 1, 2, 3, 4, 5), \tag{3}$$

the signature of a ring generated by a single degree-2 element (`scripts/validate_constraint_first.py`).

Remark 2. *The honest attribution is that this is* not new mathematics*: it is Weyl's* 1939 *theorem. The contribution here is the* identification *— that the flux pair is the $k = 2$ FFT case, so the "only exact structure is the constraint" is a corollary of classical invariant theory rather than an empirical conjecture.*

5 Mitchell's Completeness Theorem

Theorem 3 (Mitchell)**.** *Over the reals (or rationals), for $N_{\text{flux}} \neq 0$ the level set $\{F^\top JH = N\}$ is a single $\mathrm{Sp}(b_3, \mathbb{R})$ orbit. Over the integer group $\mathrm{Sp}(b_3, \mathbb{Z})$, the orbits are instead classified by the content $d = \gcd(F, H)$ ($d^2 \mid N$), so there are* finitely many *orbits per tadpole — one per divisor d of $\sqrt{N}$ — and $N = 0$ is a union of several lower-dimensional orbits.*

Sketch. Necessity is the Constraint-First Theorem: an invariant is a function of N_{flux}. Over $\mathbb{R}$, the orbit tangent space at a pair is the image of the map $\mathfrak{sp}(b_3) \to \mathbb{R}^{2b_3}$, $X \mapsto (XF, XH)$, whose rank is $2b_3 - 1$; this equals the codimension-1 dimension of the level set. The orbit is open in the level set

(equal tangent spaces), closed (an algebraic group orbit), and the level set is connected for $N \neq 0$, so the real orbit *is* the level set. Over the integers this argument does not apply (a countable orbit is never open in the real level set); the content $d = \gcd(F, H)$ is a $\mathrm{Sp}(b_3, \mathbb{Z})$-invariant with $d^2 \mid N$, so non-squarefree N splits into multiple integral orbits (Wall's theorem on extending symplectic isometries). □

The real rank is computed exactly (integer rref) at three generic pairs and at the canonical pair (e_0, e_3) itself, giving $2b_3 - 1 = 11$ for $b_3 = 6$ (`scripts/validate_no_fingerprint.py`).

Corollary 1 (No exact polynomial invariant selector)**.** *No exact, polynomial, duality-invariant function of the flux can single out a vacuum: the only exact polynomial invariant is N_{flux}, and over the integers it alone does not even distinguish the finitely many orbits sharing a tadpole (that takes the content d). This is a no-go on* exact polynomial invariant *selectors — it does* not *rule out statistical, moduli-dependent, or non-polynomial selection criteria, which the landscape program can in principle still use.*

6 Full-duality completion

Theorem 4. *The invariant ring of the* full *duality group* $\mathrm{Sp}(b_3) \times \mathrm{SL}(2, \mathbb{Z})$ *is also* $\mathbb{C}[F^\top JH]$*, so the no-go holds for the complete duality group, not just the monodromy subgroup.*

By the Constraint-First Theorem the $\mathrm{Sp}(b_3)$-invariants are $\mathbb{C}[N_{\mathrm{flux}}]$; the generator N_{flux} is $\mathrm{SL}(2, \mathbb{Z})$-invariant (verified symbolically for the generators $S : (F, H) \mapsto (-H, F)$ and $T : (F, H) \mapsto (F, F + H)$, and for a general SL(2) element $N_{\mathrm{flux}} \mapsto (ad - bc) N_{\mathrm{flux}}$), so $\mathrm{SL}(2, \mathbb{Z})$-invariance removes nothing. The full invariant ring is exactly $\mathbb{C}[N_{\mathrm{flux}}]$ (`scripts/validate_full_duality.py`).

7 The Complete Exact Structure Theorem

Theorem 5 (Complete Exact Structure)**.** *For a* CY_3 *orientifold flux compactification, every exact (moduli-independent, duality-invariant) observable is either (i) the D3 tadpole N_{flux} — the unique flux observable — or (ii) a topological invariant of the Calabi–Yau $(h^{1,1}, h^{2,1}, c_2, \kappa)$. There is nothing else.*

In particular W_0, g_s, the masses, the saxion values, the gauge couplings and the cosmological constant are *not* exact observables: they are moduli- or flux-dependent, and duality-frame-dependent, hence exactly the quantities a theory of everything does *not* get to predict.

Corollary 2 (Hard ceiling on the verification deficit)**.** *The flux sector contributes exactly* one *observable to the Mitchell Verification Deficit m. By the no-go there is provably no second flux observable to find, so m can never grow from the flux sector; future growth must come from the other sectors (D-brane charges, the Standard Model), which is the $m = 0$ frontier.*

Remark 3 (Falsifiability)**.** *The whole tower stakes itself on one checkable claim:* no moduli-independent, duality-invariant flux quantity exists that is not a function of $F^\top JH$. *A single counterexample falsifies it. That is what makes this a theorem rather than a slogan.*

8 The Five Pillars: independent foundations

The result rests not on one argument but on five independent families of theorems, each forcing a piece of the conclusion. We call them the five pillars.

pillar	forces
local	no local invariant exists — Darboux
algebraic	the only invariant is the tadpole — Weyl's FFT (the invariant ring is generated by one element)
global	what survives is global symplectic topology — Gromov non-squeezing, symplectic capacities, Elia
topological	the labels are topology — Atiyah–Singer ($N_{\text{gen}} = \lvert\chi\rvert/2$), K-theory (D-brane charges are $K(X)$ cl
counting	the numbers are theorems — Molien–Weyl + Hilbert–Serre (19 = orbit-space dimension), Morse

Darboux's theorem is the *local* pillar: a symplectic form has no local invariants, so no local geometric quantity (a symplectic analog of curvature) can distinguish one patch from another. It is the differential-geometric complement to the algebraic result, and the two are steps in one chain:

Darboux: local symplectic geometry $\to$ *the linear group* $\mathrm{Sp}(2n)$ $\quad$ *FFT:* $\mathrm{Sp}(2n)$*-invariants* $\to$ *the tadpole.*

Darboux says there is no *local* fingerprint; the FFT says there is no *duality-invariant* fingerprint either — only the pairing. The remaining pillars make the surviving structure precise: it is *global* symplectic topology (capacities, rigidity, homology-counted fixed points) and *topological* labels (the index theorem, K-theory), with the counts themselves as theorems (Molien–Weyl, Morse). Three of these yield sharper statements than "no fingerprint":

1. *The vacuum count is bounded, not fixed.* The signed count is the topological index; the unsigned count is only bounded below by the Betti numbers and otherwise depends on the flux — which has no fingerprint.

2. *The constraint is a K-theory vanishing.* D-brane charges are $K(X)$ classes, and the tadpole is "total charge $= 0$ in $K(X)$" — a topological statement deeper than the algebraic pairing.

3. *The vacuum space is a symplectic quotient.* By Marsden–Weinstein, the vacuum space is $\mu^{-1}(0)/G$: the tadpole is a moment map and the gauge quotient is symplectic reduction, whose invariant content is the moment polytope.

Exact-invariant search therefore fails for multiple independent reasons — no local invariant, no duality-invariant, a vacuum count that topology bounds but never fixes — and the surviving exact structure is exactly the Complete Exact Structure theorem's "constraints + labels."

9 Conclusion

The constraint-first conjecture is now a theorem. The flux sector's only exact structure is the D3 tadpole — a *constraint*, not a selector — and the theory of everything predicts nothing exactly except what it forbids (consistency constraints) and what it is handed (topology). No *exact, polynomial, duality-invariant* function of the flux can single out a vacuum; statistical, moduli-dependent, or non-polynomial selection is not ruled out, but any *exact polynomial invariant* selection is impossible.

Scope, named. The invariant-ring claim is proven for *polynomial* invariants (the FFT); smooth non-polynomial invariants are not covered by the symbolic proofs, though nothing suggests they exist. The "finitely many orbits per tadpole" statement rests on the content classification (Wall's theorem) over the integer group, plus the real transitivity argument (the orbit is closed; the $N \neq 0$ level set is connected), stated as such; their machine-checked (Lean) formalization is deferred to hardware. The topology list $h^{1,1}, h^{2,1}, c_2, \kappa$ is the standard complete topological data of a CY_3, taken as established mathematics.

References

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